EUGENIO KURI MUZQUIZ

Software Engineer Intern| Robotics Software Engineering Intern

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PROFESSIONAL SUMMARY

Dynamic Software Engineering Intern specializing in robotics, autonomous systems, and high-performance backend architectures with a strong foundational background in data science and applied mathematics. Proven track record in designing multi-layer collision-avoidance pipelines, real-time edge computer vision models, and automated distributed systems using Go, C++, and Python. Complements technical capabilities in ROS 2, PX4, and cloud infrastructure with high-impact communication skills validated as a featured TEDx speaker. Elevates engineering projects through dedicated technical mentorship of peer developers and structured financial leadership within large-scale student organizations.

SKILLS

C++	OpenCV	Cross-Functional Collaboration
Go (Golang)	YOLO	Problem-Solving
Python	ONNX GCP	Technical Mentorship
SQL	Firestore	Leadership
TypeScript	BigQuery	Communication
JavaScript	OpenAI API	Analytical Thinking
ROS 2	Git	Critical Thinking
PX4	Linux	Adaptability
Gazebo	Raspberry Pi	Attention to Detail
MAVLink	CM4	Time Management

PROFESSIONAL EXPERIENCE

Merlin Drones, Remote. Software Engineer — 05/2026 – 08/2026

- Designed and implemented a collision avoidance system for autonomous drone swarms in Go, improving flight safety through ROS 2, PX4, ORCA, and LiDAR integration.
- Developed an edge perception pipeline on Raspberry Pi CM4 using YOLO/ONNX and OpenCV, enabling accurate GPS localization and continuous target tracking.
- Implemented end-to-end MAVLink command workflows for mission cancellation, in-flight recovery, and transit-speed control, improving operator control and mission flexibility across the swarm platform.
- Engineered a one-command Gazebo simulation test harness for obstacle climb-over scenarios, automating validation and supporting repeatable regression testing in CI/CD pipelines.
- Developed Ground Control Station features including live swarm telemetry, map-based mission planning, USB GNSS management, and configurable failsafes, enhancing mission visibility, operator control, and flight safety.
- Resolved distributed systems failures by eliminating leader-election conflicts, race conditions, and command timeout issues, improving swarm stability and mission reliability.
- Collaborated with cross-functional engineers to integrate perception, navigation, and flight-control modules, ensuring reliable interoperability across the drone software stack.
- Optimized Go-based robotics software by identifying and resolving performance bottlenecks, improving system responsiveness during simulated and live flight operations.
- Performed integration and flight testing in Gazebo and PX4 simulations, validating new features and identifying software defects before deployment to hardware.

embeddingLABS, Remote, Software Engineer| May 2025 – August 2025

- Engineered an automated bookkeeping platform in Python that transformed invoices and bank statements into double-entry journal entries, reducing manual processing and improving bookkeeping accuracy.
- Automated client onboarding by deriving bookkeeping rules from historical ledgers with LLM-assisted fallback and validation pipelines, eliminating manual configuration and enabling the platform to scale beyond its three-customer limit.

- Replaced a vision-based bank statement extraction pipeline with a deterministic pdfplumber parser, streamlining document processing and reducing operational complexity.
- Developed responsive operator interfaces in Next.js, React, and TypeScript for policy diffing, FX management, onboarding, and PDF exports, streamlining internal workflows and improving operational efficiency.
- Integrated Firestore, BigQuery, and OpenAI APIs to build scalable data pipelines that automated financial document processing and bookkeeping workflows.
- Debugged and resolved onboarding failures, future-period generation bugs, and telemetry issues, improving application stability and ensuring a reliable user experience.
- Collaborated with cross-functional engineers to design, test, and deploy new bookkeeping features, accelerating product development and improving platform reliability.

Michigan Department of Robotics, Ann Arbor, MI, Instructional Aide| January 2025 – April 2025

- Mentored over 10 undergraduate robotics teams on system architecture, mechanical design, electronics integration, and software debugging, improving project quality and accelerating technical problem-solving.
- Developed iterative testing protocols that accelerated hardware-software debugging across 8 capstone robotics projects, improving system reliability and integration efficiency.
- Provided hands-on support with embedded systems, sensors, motor controllers, and robotics software, helping teams troubleshoot complex hardware and software issues.
- Assisted students in designing and implementing robotic systems using engineering best practices, including modular development, testing, and iterative improvement.
- Collaborated with faculty and student teams to evaluate designs, identify technical challenges, and deliver effective solutions throughout the robotics development lifecycle.

EDUCATION

Bachelor of Science in Engineering (B.S.E.), Data Science, Minor: Applied Mathematics, University of Michigan, Ann Arbor, MI|2024